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EventDrive: Event Cameras for Vision-Language Driving Intelligence

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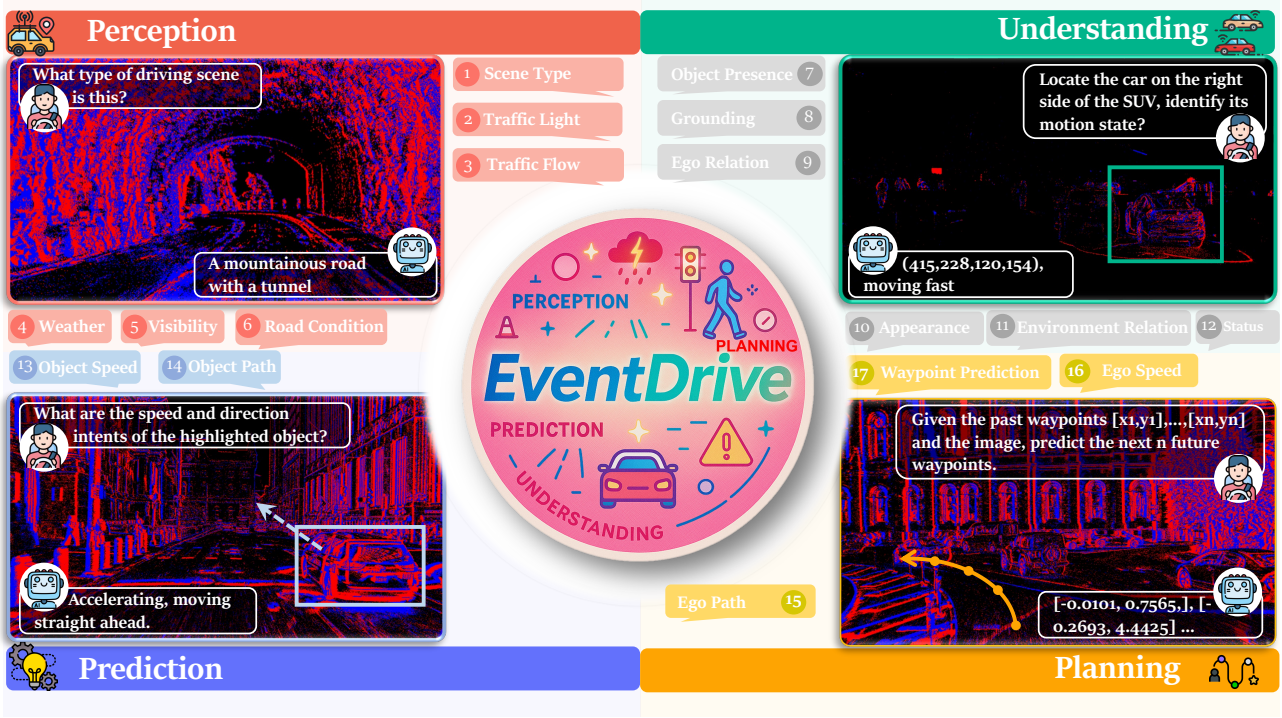


Figure 1. **Overview of the *EventDrive* benchmark.** The dataset contains **471k** event–frame–language samples across four levels of driving reasoning spanning **17** subtasks: **Perception** evaluates *scene-level context* such as scene type, traffic, and illumination. **Understanding** assesses *object-centric semantics*, including presence, motion state, and grounding. **Prediction** infers short-horizon *motion intent of surrounding agents*. **Planning** estimates *ego motion intent and future waypoints* from past trajectories. Each task is formulated through structured, language-grounded queries, enabling unified evaluation of event–frame models across the autonomy stack.

Abstract

*Event cameras sense the world through asynchronous brightness changes with microsecond latency and high dynamic range, offering motion fidelity far beyond frame-based sensors and capturing temporal structure that conventional exposures often miss. These properties make events a powerful complement to RGB in autonomous driving, especially under blur, glare, and rapid motion, where frame-based perception can become unreliable. However, existing event-aware vision–language models remain limited to generic perception and do not reveal how event sensing contributes to reasoning and decision-making across the full driving loop. We present **EventDrive**, a large-scale benchmark and model suite that unifies event streams, RGB frames, and language supervision across four core dimensions: **Perception, Understanding, Prediction, and Planning**, covering captions, structured QA, grounding, motion-state recognition, trajectory forecasting, and planning tasks. Building on this foundation, **EventDrive**-VLM introduces a multi-horizon event pyramid and a modality-routing mixture-of-experts to adaptively encode and fuse asynchronous and frame-based information for downstream reasoning. Comprehensive evaluation across diverse tasks*

shows that event streams provide substantial gains in temporal precision, motion awareness, and robustness, bringing event sensing into the center of driving intelligence.

1. Introduction

Event cameras have gained increasing attention for their ability to record dynamic scenes with microsecond temporal precision [6, 17]. Unlike RGB sensors that integrate intensity over fixed exposure periods, event cameras report asynchronous brightness changes at the pixel level [35]. This design yields high dynamic range, very low latency, and natural robustness to motion blur. These sensing advantages are especially valuable in autonomous driving, where reliable perception must be maintained across fast ego motion, rapidly moving agents, and challenging illumination conditions [10, 55]. These characteristics position event sensing as a promising complement to frame-based perception in safety-critical driving systems.

Despite this potential, events for driving research remain fragmented. Most existing efforts focus on upstream supervised tasks, *e.g.*, detection [23, 37, 48], segmentation [30, 33, 58], or optical flow estimation [64, 67]. Very few works explore high-level reasoning and decision-making that a complete driving stack requires. In contrast, the RGB community has made rapid progress toward unified vision and language models that couple perception, reasoning, and control within a single architecture [26, 27, 36, 42, 43, 61], which provides interpretability and scalable instruction-driven behavior. Early attempts to incorporate events into vision and language systems, including grounding [34] and caption-based event language models [41, 47, 65], demonstrate encouraging multimodal interactions but remain limited to generic scenes. They do not address the reasoning or decision demands that are central to real-world driving. This gap calls for a framework that integrates events throughout the autonomy pipeline in a unified and scalable manner, rather than treating it as an isolated temporal cue.

To address this need, we introduce ***EventDrive***, a unified dataset and benchmark that integrates event streams, synchronized RGB frames, and language supervision across the **full driving loop**. Our goal is to move from passive sensing to actionable understanding within a coherent multimodal interface. We decompose driving into four sequential reasoning stages: **perception**, **understanding**, **prediction**, and **planning**, each expressed as a language-grounded task probing a complementary aspect of event-based reasoning. Perception assesses robustness under challenging illumination and motion, where events offer stable edges and temporal gradients that compensate for degraded RGB signals. Understanding targets object semantics and spatial relations, with asynchronous event cues helping to disambiguate interactions. Prediction evaluates short-term behavior anticipation, where the high temporal density of events exposes velocity and acceleration. Planning examines ego intent and waypoint estimation, leveraging continuous temporal structure for steadier decisions in dynamic environments. Together, these tasks form a unified evaluation protocol that highlights how temporal cues enhance perception and reasoning across the driving stack while remaining compatible with classical vision models and modern vision–language architectures.

Building upon ***EventDrive***, we develop ***EventDrive*-VLM**, an event-driven vision–language model that integrates asynchronous event cues into unified multimodal reasoning. The framework tackles two core challenges in event-based driving. First, event streams vary widely in temporal density. To capture motion across scales, we introduce a **Dynamic Horizon Encoding module** that voxelizes events at multiple temporal resolutions and selects the most informative representation via a *mixture-of-experts* gate. This preserves high-frequency dynamics during fast motion while ensuring stable aggregation in low-motion regimes. Second, event features must be aligned with the LLM's semantic space. We address this with an **Event Q-Former Alignment module** that performs cross-attention between learnable event queries and temporally encoded representations, enabling motion-focused signals to integrate cleanly with language reasoning. A lightweight two-stage curriculum first learns event–language alignment with the LLM frozen, then performs instruction tuning to produce a coherent event-driven perception-to-action pipeline.

Comprehensive experiments on ***EventDrive*** reveal strong modality contrasts. Frame-only VLMs perform well in perception but degrade sharply under low light and motion blur, and show limited spatial reasoning and weak motion inference. Event-only models excel in speed and direction prediction, highlighting the value of high-frequency temporal cues, yet lack semantic richness for appearance-heavy understanding. Our event–frame fusion model improves performance across all task families, stabilizing perception under adverse conditions, enhancing grounding and relational reasoning, and delivering stronger motion prediction and ego intent estimation. These results show that events and frames provide complementary strengths, and their integration yields more reliable multimodal reasoning than either alone.

In summary, our main contributions are as follows:

- We introduce ***EventDrive***, the first full-stack event and language benchmark for autonomous driving that unifies perception, understanding, prediction, and planning within a consistent multimodal framework.
- We propose ***EventDrive*-VLM**, a general training framework that equips large vision and language models with the ability to interpret, align, and reason over asynchronous event representations.

- We establish a comprehensive evaluation protocol and extensively analyze how temporal cues improve multimodal reasoning, offering a foundation for future event-driven intelligence in real-world driving.

2. Related Work

Event Cameras for Driving. Event sensors offer microsecond latency, high dynamic range, and robustness to motion blur, making them ideal for driving perception. They have been applied to detection [23, 37, 48, 51, 63], segmentation [30, 33, 58], tracking [64], and recognition [13, 20, 31]. Detection methods include graph- or spike-centric models preserving sparsity and asynchrony [14, 15, 18, 50, 53, 56, 64], and dense feed-forward models that convert events into voxel grids or time surfaces for image-based backbones [9, 23, 28, 29, 39, 51, 52]. Beyond detection, event-based segmentation and tracking emphasize temporal continuity and motion cues [21, 34, 58]. However, most systems remain limited to low-level perception without connecting to higher-level reasoning and planning. Our work advances this direction by integrating event sensing into a unified framework that links perception, understanding, and decision-making for end-to-end driving intelligence.

Event-Frame Multimodal Learning. Events and RGB frames are complementary: events capture fine temporal dynamics under extreme lighting, while frames provide rich semantic context. Early pipelines relied on late fusion [8, 38], whereas later designs introduced shared backbones and attention-based gating for cross-modal interaction [4, 48, 59]. Transformer-based approaches further improved temporal reasoning and adaptive weighting under complex motion and illumination [37, 62, 66], benefiting tasks such as deblurring, depth estimation, and segmentation [22, 25, 57]. Yet most frameworks rely on fixed temporal windows, which fail to capture motion cues that unfold over varying temporal scales [5, 19, 37]. Our method addresses this by aligning multi-horizon event streams with frame semantics, enabling frequency-adaptive fusion that maintains both temporal precision and semantic coherence.

Event-based VLMs. While vision-language models (VLMs) have transformed visual understanding, extending them to asynchronous event data remains challenging [33, 34]. The key difficulty lies in embedding sparse, high-temporal-resolution signals into a shared vision-language space. Early contrastive approaches adapt CLIP-style alignment for zero-shot recognition [33, 62], but rely on limited-scale datasets and rasterized inputs, reducing temporal fidelity. Recent works narrow this gap: EventGPT [47] couples an event encoder and temporal aggregator with an LLM; EventVL [41] trains on large-scale event-image-text pairs with spatiotemporal and semantic alignment; and LLaFEA [65] integrates frame-event fusion with cross-attention and duration cues. Despite these advances, most models remain confined to captioning or short QA tasks. On the contrary, our method preserves temporal sparsity and extends event-based VLMs toward comprehensive driving reasoning across diverse tasks.

3. *EventDrive*: A Vision–Language Benchmark for Event-Based Driving

In this section, we present the ***EventDrive*** dataset and benchmark in detail. Sec. 3.1 introduces the overall motivation and task hierarchy that define the dataset design. Sec. 3.2 describes the annotation pipeline and data composition, highlighting how multimodal inputs are processed into language-grounded supervision. Sec. 3.3 summarizes the data splits and benchmark statistics, outlining the scale and coverage of the dataset and its comparison with existing event-based vision–language benchmarks.

3.1. Hierarchical Task Framework

Event cameras remain reliable under low light and rapid motion, offering advantages for safety-critical driving where conventional frames often fail. Existing event-based works focus mainly on isolated perception tasks, while RGB-based methods have progressed toward unified vision–language–action frameworks. To close this gap, we build a dataset that extends event understanding across the entire autonomy stack. The driving loop is organized into four sequential steps: environmental **perception**, object-centric **understanding**, forward **prediction**, and ego-driven **planning**. Each step is formulated as a language-grounded task that aligns event representations with multimodal models, enabling a systematic evaluation of how event signals contribute to high-level reasoning.

Perception describes global scene conditions across six subtasks: [1]*scene type*, [2]*visibility*, [3]*traffic flow*, [4]*weather*, [5]*traffic light*, and [6]*road condition*. Captions and structured QA pairs assess whether models can interpret environmental context from short temporal windows. Event streams preserve edge contrast and temporal gradients under challenging illumination or motion blur, strengthening robustness beyond frame-based perception.

Understanding captures object-level semantics and spatial structure using captions, QA pairs, and bounding boxes spanning [1]*object presence*, [2]*appearance*, [3]*motion state*, [4]*ego relation*, [5]*environment relation*, and [6]*grounding*. The asynchronous event signal supplies fine temporal cues that support reasoning about motion and spatial relations that static frames often miss.

Prediction targets short-horizon behavioral forecasting through indicators of [1]*speed* change and [2]*direction* change. Each instance uses language queries to evaluate whether models can infer future motion trends of surrounding objects from limited observations. The high temporal density

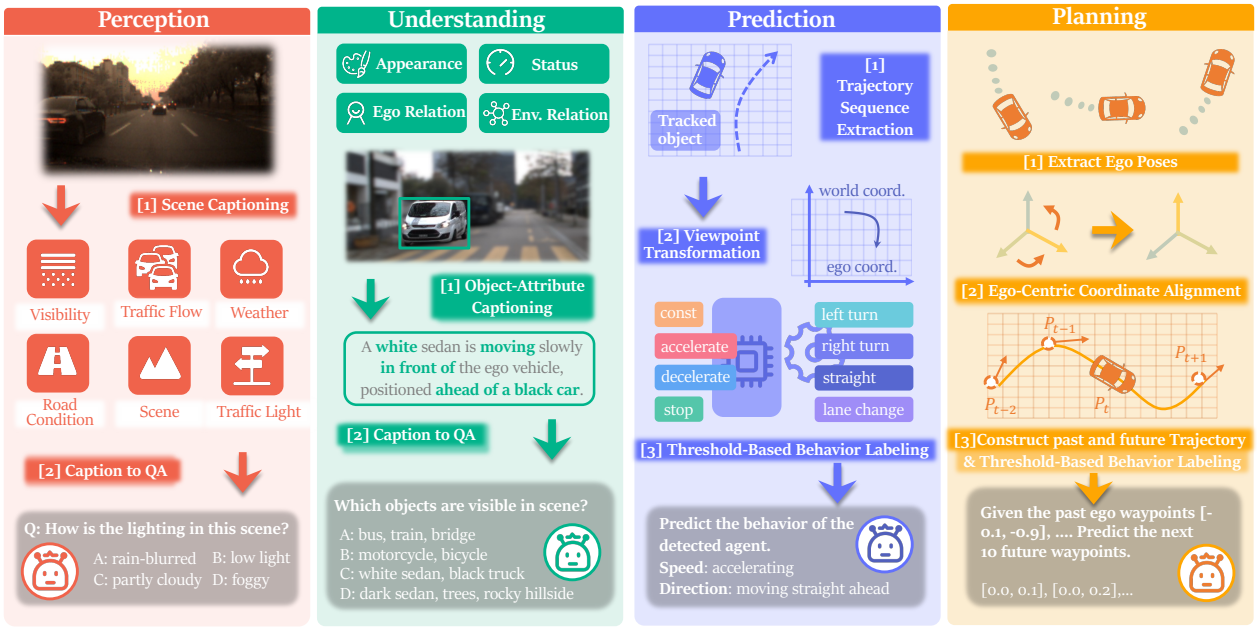


Figure 2. Annotation pipelines of ***EventDrive***. **Perception** converts scene-level attributes into structured QA; **Understanding** generates object-level semantic captions and transforms them into QA; **Prediction** extracts trajectories, applies ego-frame transformation, and assigns motion labels; **Planning** constructs ego-centric waypoints and produces corresponding decision-oriented supervision.

of events reveals velocity and acceleration directly, improving the fidelity of motion forecasting.

Planning evaluates ego-centric decision making using [1]*speed intent*, [2]*direction intent*, and [3]*waypoint planning*. Models must convert multimodal observations into coherent driving decisions. Event sensing preserves continuous awareness of dynamic changes, supporting more stable reasoning in rapidly evolving or low-visibility conditions.

3.2. Language-Grounded Data Generation

We construct ***EventDrive*** through a semi-automatic pipeline built upon the event datasets DSEC [24], M3ED [7], and PKU-DAVIS-SOD [37], which span diverse environments and illumination conditions. Using synchronized RGB frames, event streams, bounding boxes, LiDAR, and ego-pose signals, we employ Qwen3-VL [2] to generate linguistically structured supervision across the four task dimensions. The pipeline, shown in Fig. 2, integrates automatic captioning with controlled question–answer generation to ensure both scalability and semantic consistency.

Scene-level **perception** labels are produced by prompting Qwen3-VL to generate global captions describing the six environmental attributes, which are subsequently decomposed into balanced question–answer pairs with explanatory sentences. Object-level **understanding** builds on ground-truth bounding boxes from DSEC. Qwen3-VL first generates descriptions for the five object attributes, and these descriptions are transformed into visual question–answer and grounding tasks that link textual queries

Table 1. Statistics across four driving reasoning tasks.

Task	Source	#Train	#Test	#Hard
Perception	DSEC [24]	53,196	12,828	780
	M3ED [7]	65,280	32,136	15,396
	PKU [37]	46,860	11,400	3,960
Understanding	DSEC [24]	91,716	28,848	1,812
Prediction	M3ED [7]	7,784	2,810	1,970
Planning	M3ED [7]	46,290	29,526	18,951
Total		**311,126**	**117,548**	**42,869**
Grand Total Samples			**471,543**	

to specific spatial regions. For **prediction**, we use ego-pose to project 3D boxes from [44] into the ego frame, extract object trajectories, and convert their motion patterns into natural-language descriptions of speed and direction intent, paired with corresponding QA items. Ego-level **planning** leverages M3ED trajectory supervision to derive speed intent, path intent, and future waypoints, which are aligned with language queries that evaluate ego decision making. Together, these components yield a unified annotation framework that connects perception, understanding, prediction, and planning through temporally and semantically coherent language-grounded supervision.

3.3. Compositional Structure and Statistics

We adopt standard *training* and *testing* splits and further introduce a **hard** split consisting solely of *low-light* and *motion-blur* sequences. This split enables targeted evaluation of the advantages of event sensing under conditions where frame-based perception degrades. As shown

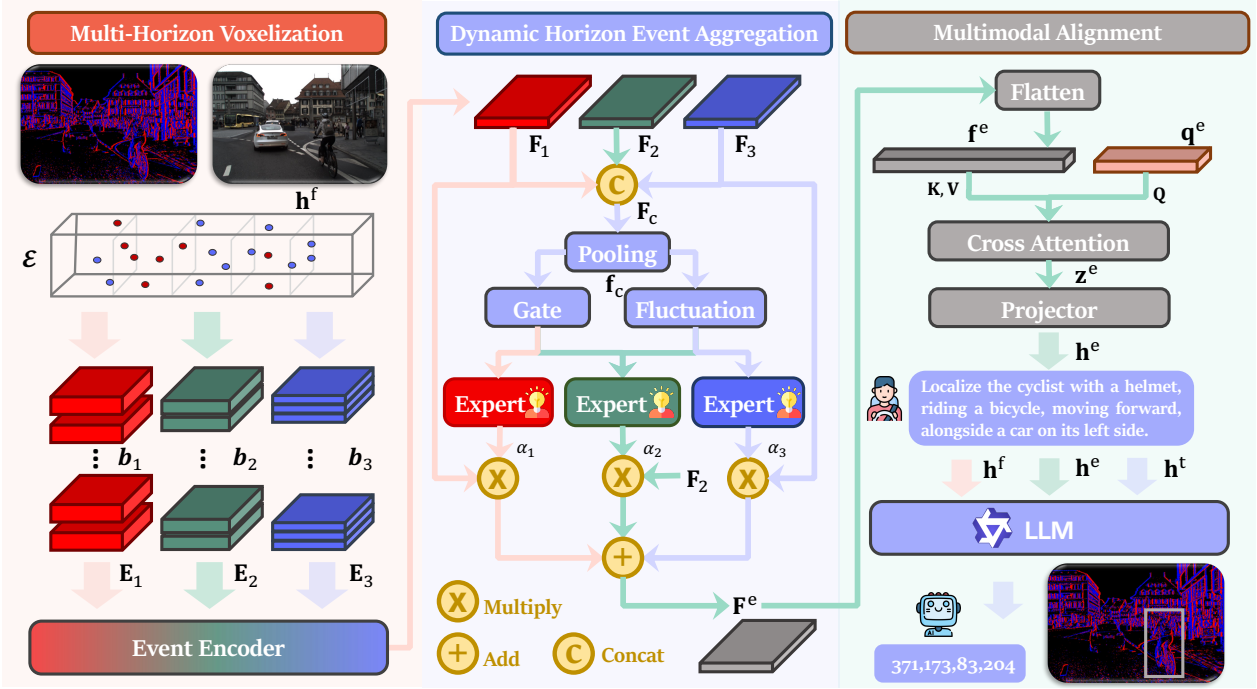


Figure 3. ***EventDrive*-VLM Overview.** We first convert asynchronous events into **multi-horizon voxel tensors** that capture motion at *different temporal scales*. A **dynamic horizon event encoder** then aggregates these representations through a *Mixture-of-Experts* gating mechanism (*cf*. Sec. 4.1). An **Event Q-Former** performs cross-attention to extract language-aligned, motion-aware tokens, enabling coherent fusion of multi-modal features within the LLM for unified driving reasoning (*cf*. Sec. 4.2). A **two-stage training curriculum** further aligns event, visual, and linguistic pathways, strengthening cross-modal grounding and downstream reasoning performance (Sec. 4.3).

in Tab. 1, the dataset contains **471,543** event–frame–text samples, offering large-scale multimodal supervision across perception, understanding, prediction, and planning. Compared with existing event–language datasets [41, 47, 65], which often rely on simulated data or provide limited real-world coverage below 100k samples, ***EventDrive*** offers a significantly broader and more diverse set of real-world annotations. It supports a wider range of tasks that connect event-driven perception with higher-level reasoning and decision-making, forming a more comprehensive benchmark for evaluating multimodal driving intelligence.

4. *EventDrive*-VLM: A Unified Model for Event-Based Driving Intelligence

Building upon ***EventDrive***, our goal is to develop an event-driven vision–language–action model that can interpret and generate decisions grounded in the spatio-temporal structure of event streams. As shown in Fig. 3, the core of ***EventDrive*-VLM** is a multimodal LLM backbone that integrates asynchronous event cues with pre-trained visual–linguistic knowledge to support coherent reasoning across driving tasks. To bridge event dynamics with the LLM embedding space, the model incorporates three components: a **dynamic horizon event encoder** that adapts to varying sampling frequencies and motion patterns (*cf*. Sec. 4.1), an **Event Q-Former** that extracts language-aligned and motion-aware representations (*cf*. Sec. 4.2), and a **two-stage training curriculum** that progressively aligns event, visual, and linguistic pathways for unified multimodal reasoning (*cf*. Sec. 4.3).

4.1. Dynamic Horizon Event Encoder

Event data exhibit large variation in temporal density across datasets and tasks. Sensors operate at different sampling rates, and the required temporal resolution also depends on the objective: perception benefits from broader temporal context, whereas prediction and planning rely on fine-grained motion cues. Conventional voxelization [23] applies a fixed number of temporal bins and thus compresses long exposure windows while blurring fast motion, losing the high-frequency detail essential for motion reasoning.

To address this issue, we propose a **dynamic horizon encoding** strategy that adapts to varying time scales and scene dynamics. Given an event stream $\mathcal{E} = \{e_k\}_{k=1}^{K}$, where each event $e_k = (x_k, y_k, t_k, p_k)$ encodes spatial coordinates, timestamp, and polarity $p_k \in \{-1, 1\}$, standard voxelization maps events into a 4D tensor $\mathbf{E} \in \mathbb{R}^{2\times B\times H\times W}$:

$$\mathbf{E}(p,\tau,x,y) = \sum_{e_k\in\mathcal{E}} \delta(p-p_k)\delta(x-x_k,y-y_k)\delta(\tau-\tau_k), \quad (1)$$

where $\tau_k = \left\lfloor \frac{t_k - t_a}{t_b - t_a} B \right\rfloor$. Instead of relying on a single bin size B, we construct multiple voxel tensors using temporal resolutions $\mathcal{B} = \{b_n\}_{n=1}^{N}$, producing event tensors $\mathbf{E}_n$ that capture short-, medium-, and long-horizon motion patterns.

To adaptively select the most suitable temporal horizon for different motion patterns and task requirements, we employ a *Mixture-of-Experts (MoE)* [54] gating mechanism that dynamically weights multi-scale temporal experts according to scene dynamics. Each expert network [23] specializes in one temporal resolution, processing a voxelized tensor $\mathbf{E}_n$ with bin size b_n to produce an encoded feature

$$\mathbf{F}_n = \boldsymbol{\sigma}(\mathbf{E}_n), \qquad \mathbf{F}_n \in \mathbb{R}^{H \times W \times d}, \tag{2}$$

where d is the channel dimension. The resulting expert features are concatenated into a contextual representation $\mathbf{F}_c \in \mathbb{R}^{H \times W \times (Nd)}$. Global average pooling produces a compact descriptor $\mathbf{f}_c \in \mathbb{R}^{Nd}$ that summarizes multiscale temporal cues. The gating logits are then computed as

$$\mathbf{z} = \mathbf{W}_g \mathbf{f}_c + \texttt{Softplus}\big(\boldsymbol{\epsilon} \odot (\mathbf{W}_{\text{noise}} \mathbf{f}_c)\big), \tag{3}$$

where $\mathbf{W}_g, \mathbf{W}_{\text{noise}} \in \mathbb{R}^{Nd \times N}$ are trainable parameters, and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, 1)$ introduces controlled stochasticity to encourage expert diversity. Retaining only the largest k logits focuses computation on the most relevant temporal experts, and softmax normalization yields the weights α_n. The aggregated event representation is computed as $\mathbf{F}^e = \sum_{n=1}^{N} \alpha_n \mathbf{F}_n$. This adaptive formulation allows the encoder to emphasize high-resolution temporal features when motion is fast and to leverage coarse but stable aggregation when dynamics are mild. As a result, the model maintains temporal fidelity across a wide range of driving scenarios while remaining computationally efficient.

4.2. Event Q-Former Alignment

After obtaining the aggregated event representation $\mathbf{F}^e$, we align it with the LLM embedding space using an **Event Q-Former (EQA)**. Rather than concatenating event and frame tokens, which ignores modality asymmetry and leads to high computational cost, the EQA employs cross-attention to extract the motion-relevant components of the event representation that matter for language-guided reasoning.

Following the Q-Former architecture [40], we introduce a set of learnable event query tokens $\mathbf{q}^e \in \mathbb{R}^{N_q \times d}$, which attend to the event feature map $\mathbf{F}^e \in \mathbb{R}^{H \times W \times d}$. After flattening $\mathbf{F}^e$ into $\mathbf{f}^e \in \mathbb{R}^{(HW) \times d}$, the attended event embeddings are computed as

$$\mathbf{z}^e = \texttt{softmax}\left(\frac{(\mathbf{q}^e \mathbf{W}_Q)(\mathbf{f}^e \mathbf{W}_K)^\top}{\sqrt{d}}\right)(\mathbf{f}^e \mathbf{W}_V), \tag{4}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d}$ are learnable projections. The output $\mathbf{z}^e \in \mathbb{R}^{N_q \times d}$ forms a compact representation that summarizes the most salient temporal and motion cues.

The Event Q-Former enables each query to attend selectively to temporally informative regions in the event stream, producing motion-aware embeddings suitable for downstream multimodal reasoning. This structured attention preserves the temporal distinctiveness of events while supporting coherent interaction with frame and text representations. A lightweight projection layer maps the attended event features into the LLM embedding space, yielding event tokens $\mathbf{h}^e$ that align with the projected frame tokens $\mathbf{h}^f$ and text embeddings $\mathbf{h}^t$. The concatenated sequence is then used as the input to the LLM decoder, enabling unified multimodal reasoning across all driving tasks.

4.3. Training Curriculum

We adopt a two-stage training curriculum that progresses from event–language grounding to multimodal instruction following, enabling stable and efficient adaptation of the event-driven VLM.

Event–Language Pre-adaptation. In the first stage, the LLM and the frame visual encoder are kept frozen, while the event encoder, Q-Former, and projection layers are trained with a language modeling objective on paired caption data. Although frame data are available, gradients flow only through the event pathway, allowing the model to form a linguistically aligned event representation without altering the pretrained frame semantics. This generative supervision encourages the event encoder and Q-Former to organize temporal cues and motion structure into embeddings compatible with the LLM space, providing a stable form of cross-modal alignment.

Instruction Tuning. In the second stage, we unfreeze the transformer blocks of the LLM and fine-tune the entire event pathway together with these LLM layers on all caption and QA data, while keeping the frame visual encoder frozen. This phase integrates temporal and semantic signals more tightly, enabling consistent reasoning across perception, understanding, prediction, and planning. Joint optimization under multimodal instructions yields coherent grounding between event dynamics and textual responses, forming a unified event-driven perception-to-action model.

5. Experiment

5.1. Experimental Settings

Implementation Details. We fine-tune Qwen2.5-VL-7B-Instruct [3] with a pretrained RVT [23] backbone as the event encoder. Dynamic horizon encoding uses temporal bins $\mathcal{B} = 20, 50, 100$, and the event projector is a linear layer. The RGB visual tower remains frozen. Training adopts AdamW [32] with a cosine schedule on 16 NVIDIA H20 GPUs, using a learning rate of 1×10^{-4} for the event encoder and projector. Mixed precision bf16 is used throughout. Pre-adaptation and instruction tuning each run

Table 2. **Comparison across four driving–reasoning tasks** in the ***EventDrive*** benchmark. For **perception**, we report `QA Accuracy` on three subsets. For **understanding**, we report `QA Accuracy`, grounding Top-1 `Accuracy` at IoU 0.6, and **mIoU**. For **prediction** and **planning**, we report `Speed Accuracy` and `Path Accuracy`, and for planning, we also report the mean **L2 Error** (lower is better). All scores are reported in percentage (%), except for L2 Error in meters. Best results are shown in **bold** and 2nd-best results are underlined.

Method	Perception			Understanding			Prediction		Planning		
	Acc@D	Acc@M	Acc@P	Acc	Acc@60	mIoU	Speed	Path	Speed	Path	L2 Error
Event-based Models											
EventGPT-7B [47]	51.40	54.97	52.25	38.78	5.49	8.24	27.84	65.44	28.99	76.08	11.42
EventDrive-VLM	62.51	67.07	62.39	54.21	43.43	47.82	34.68	82.25	42.56	84.64	6.89
Frame-based Models											
LLaVA-v1.6-Mistral-7B-hf [45]	58.65	59.24	50.05	40.37	15.02	29.0	18.22	29.68	9.89	87.51	8.13
LLaVA-OneVision-1.5-8B [1]	80.81	86.35	72.42	61.62	2.91	12.5	14.38	83.91	56.51	58.50	9.19
InternVL2.5-8B [11]	78.42	83.41	75.86	60.02	0.29	2.5	39.72	86.41	12.52	73.34	10.19
InternVL3-8B [68]	78.61	86.27	74.37	60.60	0.24	2.31	4.41	**91.25**	52.04	84.34	9.84
Qwen2.5-VL-7B-Instruct [3]	77.71	76.92	62.46	49.98	63.35	60.11	8.47	81.64	39.32	87.41	8.78
Qwen2.5-VL-7B-Instruct* [3]	81.52	84.69	75.88	58.44	69.94	67.19	36.84	84.94	53.87	89.44	4.54
Event + Frame Models											
EventDrive-VLM	**85.44**	**86.64**	**78.89**	**65.46**	**72.86**	**72.56**	**42.44**	87.49	**57.03**	**92.35**	**3.66**

Table 3. **Comparative results on the Event-Chat dataset [47]**, which evaluates event-driven description/reasoning through **Detailed Captioning**, **Complex Reasoning**, and **Visual Question Answering**. Higher values indicate better performance.

Models	Params	DC	CR	VQA
Frame-based Models				
LLaVA-7B-v1.5 [46]	7B	2.20	4.04	3.26
Qwen2-VL-7B [60]	7B	2.38	4.02	2.91
InternVL2-8B [12]	8B	2.37	4.00	3.71
Deepseek-vl-7b [49]	7B	2.41	4.10	3.37
Event-based Models				
EventGPT-7B [47]	7B	**3.52**	4.09	**4.29**
EventDrive-VLM	7B	3.43	**4.15**	3.94

for two epochs with a batch size of 128. Packed sequences and flattened multimodal inputs are applied with a sequence length of 4096 tokens, and FlashAttention 2 [16] accelerates all attention layers.

Evaluation Metrics. We establish a unified evaluation protocol across all defined tasks. Perception is evaluated by QA accuracy over six scene attributes. Understanding combines QA accuracy across five object-level aspects with grounding metrics, including Top-1 localization accuracy at IoU 0.6 and mean IoU. Prediction assesses speed and path intent accuracy. Planning evaluates intent accuracy together with mean L2 waypoint error across 1, 3, and 5 seconds. Additional details appear in the supplementary materials.

Baselines. Comparisons include both *frame-only* and *event-only* models. Frame-only evaluation covers Qwen2.5 VL [3], InternVL [11, 68], and LLaVA [1, 45] under zero-shot inference, which tests robustness in low light and motion-blurred scenes. To isolate the contribution of events, we additionally fine-tune a Qwen2.5-VL-7B model using the same instruction tuning protocol as our model. For event-only comparison, we evaluate the open-sourced EventGPT in a zero-shot manner since its training code is unavailable. We further report zero-shot results on the released portion of the Event-Chat [47] dataset, noting that the absence of an official split limits this evaluation to reference use.

5.2. Comparative Studies

Comparisons with Event-Only Methods. Event-only models in Tab. 2 perform well on motion-centric tasks such as speed and path intent, confirming that high-frequency temporal cues are inherently well suited for short-horizon motion reasoning. They also show reasonable grounding performance despite lacking explicit spatial supervision, highlighting the structural information encoded in event streams. Yet, the absence of appearance cues limits their semantic understanding, resulting in lower perception and object-centric accuracy. Results on the Event-Chat benchmark (Tab. 3) further show that the representations learned from ***EventDrive*** transfer across datasets, indicating that the dataset promotes generalizable event–language alignment rather than overfitting to a specific annotation style.

Comparisons with Frame-Only Methods. Frame-based VLMs achieve high accuracy on perception under normal conditions but degrade sharply in `Acc@P`, understanding `Acc@60`, and speed-related prediction metrics due to their inability to encode motion or handle low-light and blur. These limitations lead to unstable spatial reasoning and poor motion intent inference. Introducing events alleviates these issues: event–frame fusion improves all metrics, especially grounding and speed intent, showing that temporal gradients and motion cues are essential for reliable reasoning and cannot be inferred from RGB imagery alone.

Comparisons across Different Tasks. The four task families reveal complementary strengths of different modalities. Perception is broadly solvable, but event streams offer improved robustness under visual degradations. Un-

Table 4. Ablation results on the ***EventDrive*** benchmark. We report average `QA Accuracy` for perception, `QA Accuracy` and **mIoU** for understanding, and mean `Speed/Path Accuracy` for prediction and planning. Planning additionally includes mean **L2 Error** in meters (lower is better).

Method	Per.	Und.		Pre.	Plan.	
	Acc	Acc	mIoU	Acc	Acc	L2 Error
Voxelization						
N = 1	82.40	62.33	69.52	63.96	71.18	4.11
N = 5	**83.95**	64.97	72.25	62.78	73.49	3.88
Dynamic Horizon Event Aggregation						
Add	76.76	63.89	67.64	62.50	72.49	4.57
Wt.sum	83.84	64.67	70.56	61.08	74.10	3.75
Multimodal Alignments						
Concat	79.35	61.08	71.93	64.76	72.62	4.01
Attention	81.25	65.12	70.23	62.14	**75.85**	3.69
Ours	83.66	**65.46**	**72.56**	**64.96**	74.69	**3.66**

derstanding exposes significant variance across VLMs, as many struggle with spatial relations and grounding from a single RGB frame; temporal cues help resolve these ambiguities. Prediction presents the largest modality gap: inferring speed or direction from static frames is ill-posed, whereas events directly encode motion, leading to consistently higher accuracy for event-enhanced models. Planning mirrors this pattern, where temporal dynamics from traffic improve speed intent estimation and reduce trajectory L2 error. These trends collectively demonstrate that events provide structural and temporal information that complements semantics from frames across the full driving chain. A qualitative comparison is shown in Fig. 4.

5.3. Ablation Studies

Multi-Horizon Voxelization. We evaluate the impact of using different numbers of temporal resolutions when voxelizing event streams. As shown in Tab. 4, increasing the number of horizons from one to five improves performance across all tasks, particularly in understanding and planning, where temporal diversity helps disambiguate object motion and spatial relations. However, the performance gain saturates beyond three horizons, while computational cost grows proportionally with the number of voxelized tensors. Our choice ($N = 3$), therefore, provides the best trade-off between accuracy and efficiency, capturing both short-range motion cues and longer-term temporal context without incurring unnecessary overhead.

Dynamic Horizon Event Aggregation. We compare different strategies for aggregating multi-horizon event features. A naïve summation ("Add") collapses temporal distinctions and yields the weakest results, confirming that indiscriminate fusion fails to preserve horizon-specific information. Weighted summation ("Wt.sum") improves performance by allowing the model to favor informative horizons, but still underperforms compared with selecting a single expert. Our MoE-based dynamic horizon module, which activates only the top-scoring expert, achieves the best overall results, indicating that event representations benefit from horizon specialization and that suppressing irrelevant resolutions is more effective than blending them.



Figure 4. Qualitative results on ***EventDrive*** comparing ***EventDrive***-VLM with Qwen. Events remain reliable under low light and motion, improving scene perception, object understanding, motion prediction, and ego intent estimation.

Multimodal Alignment. We study the alignment between event features and the multimodal embedding space. Simple concatenation of event and frame tokens leads to suboptimal performance due to modality imbalance and increased sequence length. Cross-attention improves grounding and planning by enabling token-level interactions, yet remains inferior to our Event Q-Former. The Q-Former achieves the best results while maintaining a lower computational cost, as learnable queries extract only the most salient motion patterns rather than attending to the full spatiotemporal map. This confirms that structured, query-centric alignment offers both efficiency and stronger motion abstraction.

6. Conclusion

In this work, we introduce ***EventDrive*** and ***EventDrive***-VLM to unify event sensing, frames, and language across the full driving stack. Our results show that high-frequency temporal cues provide complementary strengths to RGB and significantly enhance multimodal perception, reasoning, and control. Event–frame fusion improves robustness and motion understanding, highlighting events as a key modality for future driving systems.

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